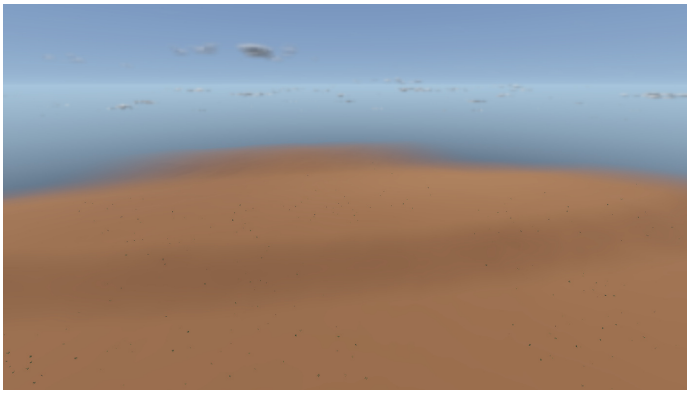
# Regional landscapes

The same broad biome can now contain different ground, relief, vegetation and water settings.

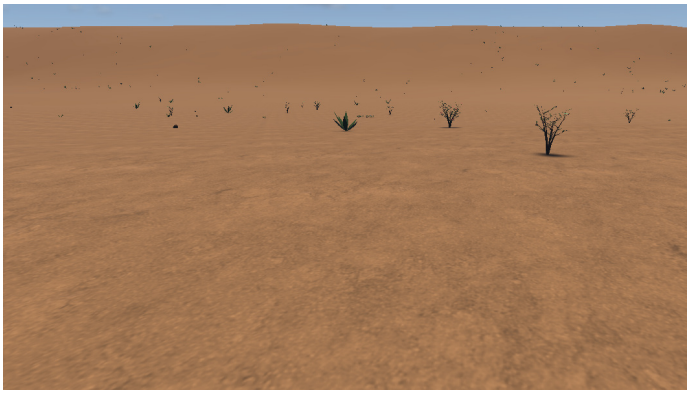
Update 2 adds local landscape character inside the existing world: sandy and gravelly deserts, dry basins, different wetlands, rocky uplands and enclosed ponds. The twenty images below are real wide/ground captures of ten sampled environments.

## Sand seas

Loose sand supports dune relief and a sandy surface, with sparse dryland plants instead of a grass lawn. Relief looks modest in this midday view; ripples and surface texture are more apparent up close.



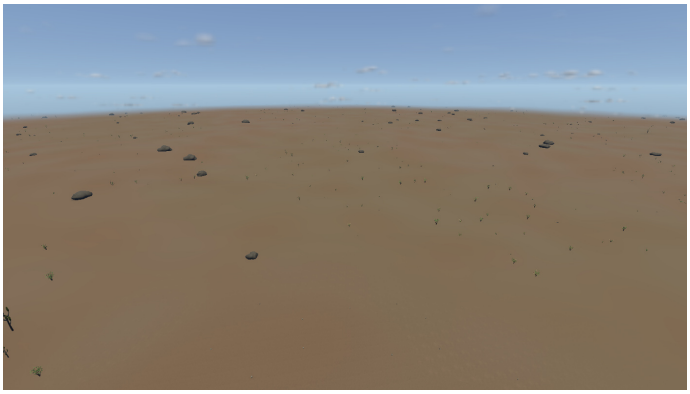
Wide view



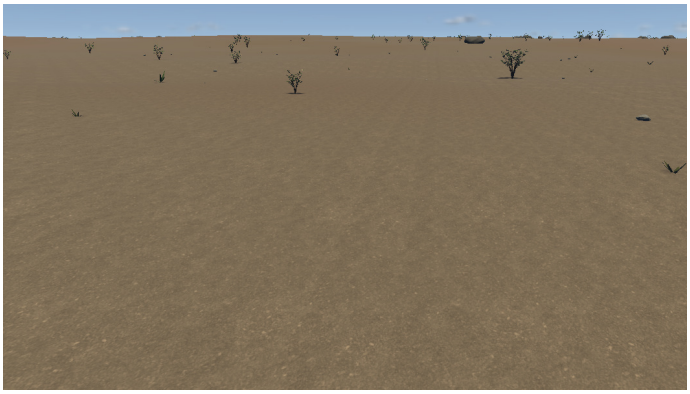
Ground detail (crop)

## Gravel deserts

A firmer gravelly surface carries scattered coarse stones and sparse scrub. This is a different substrate from mobile sand, rather than the same desert ground with a different label.



Wide view



Ground detail (crop)

## How regions are chosen

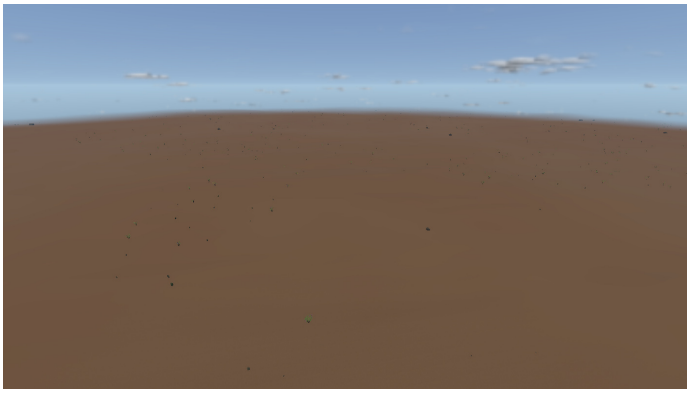
Soil, rock, sediment, moisture, drainage, canopy and cold-ground conditions feed blended local controls. This changes physical ground and plant placement, not just the displayed biome name.

# Dry basins: clay and salt

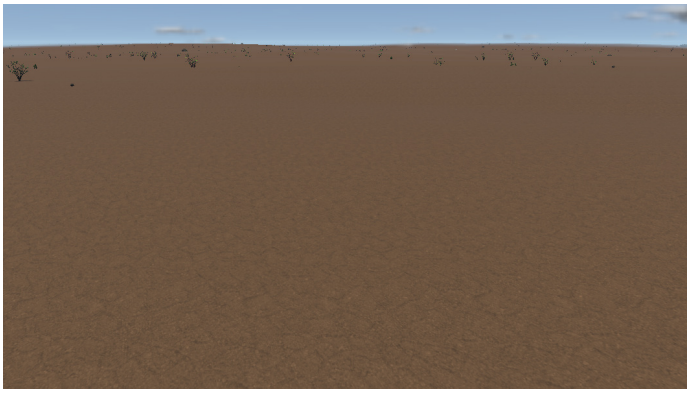
Flat dry landscapes now differ in their surface material and ability to support plants.

## Clay pans

Fine, dry basin ground uses a darker cracked-mud surface and sparse scrub. The cracking is small-scale material detail; the wide view remains a broad, relatively flat basin.



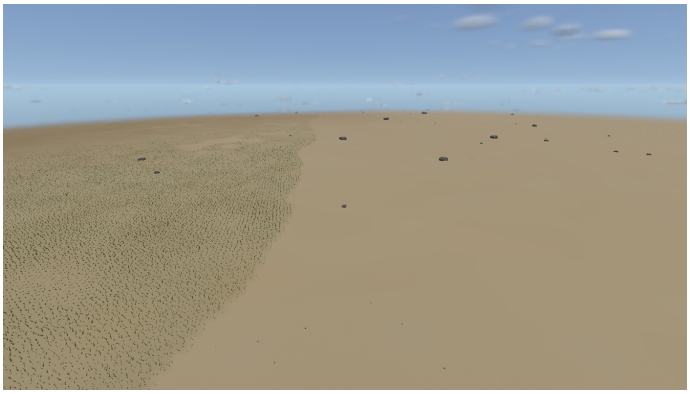
Wide view



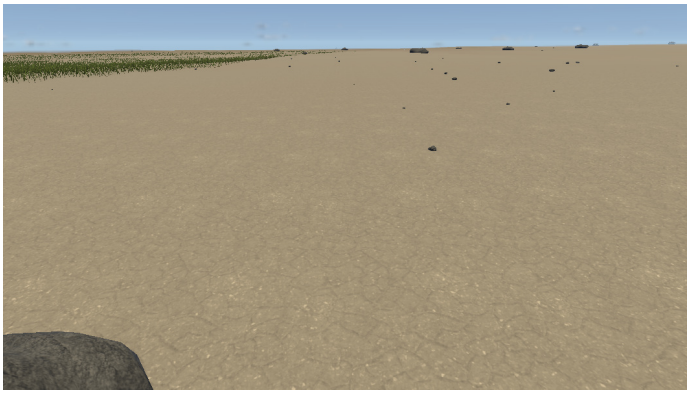
Ground detail (crop)

## Salt flats

Pale crust and fine drying cracks contrast with the clay pan. Cover is sparse on the flat, with more vegetation toward the less saline margin; this example is not a white mirror-like salt lake.



Wide view



Ground detail (crop)

## More than these four dryland examples

Related controls also cover sand sheets, exposed rock deserts, loess hills and dissected soft ground. These ten sampled scenes are not an exhaustive showcase of every landscape label or every environment on Earth.

## New ground surfaces

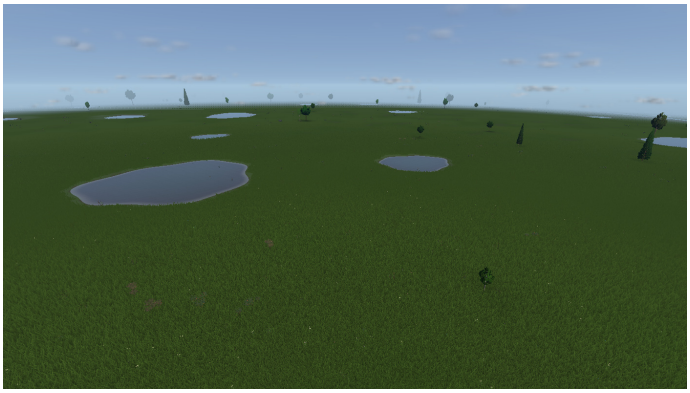
Photographed sand, gravel and cracked mud replace four existing surface slots. The ground texture arrays retain the same fourteen-layer size and GPU payload; there are no extra ground-rendering passes for these surfaces.

# Open marsh to wooded wetland

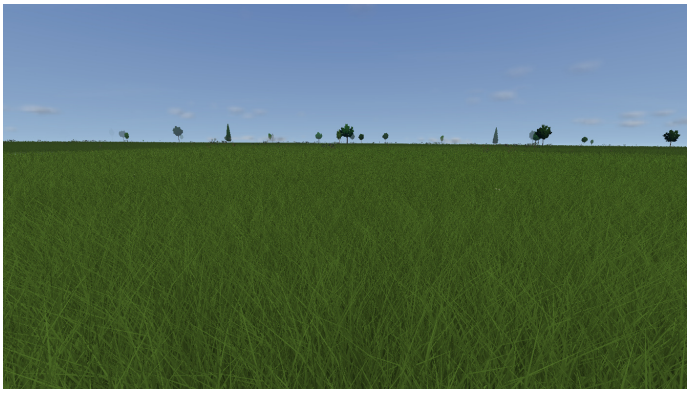
Wet ground should not always turn into either an ordinary forest or a uniform open lawn.

## Freshwater marsh

Open sedge/rush cover surrounds shallow pools, with very little woody cover at this site. The neighbourhood contains three pools within 250 m; twenty nearby reed plants are rooted in shallow water.



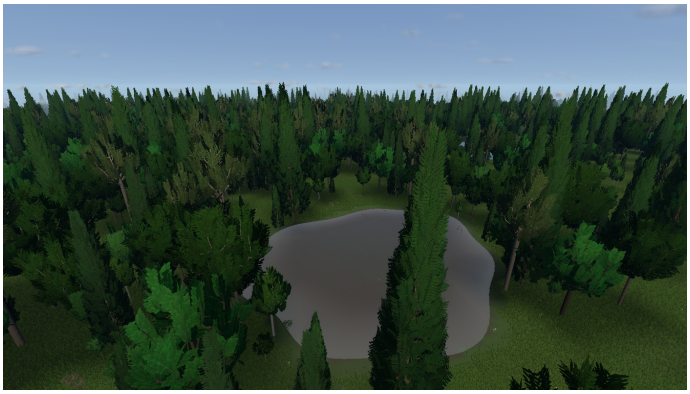
Wide view



Ground detail

## Wooded wetland

Suitable saturated ground can retain a forest around standing water. This site has four nearby pools and thirty-six reeds rooted in water. It is an actual wooded/water setting, rather than a moist upland forest renamed as swamp.



Wide view



Ground detail

## Vegetation follows the surface

Wetness and rooting conditions affect tree admission. Reed-form plants can occupy shallow, slow water; ordinary shrubs stay on land. Cold, saline, shallow-soil and persistently flooded sites retain their exclusions.

## What still looks basic

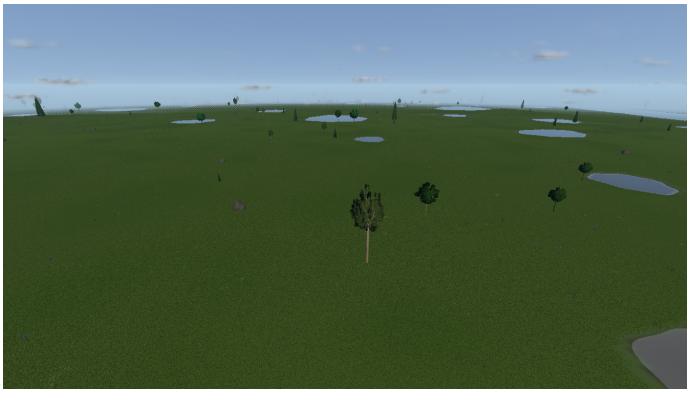
The cover mix is functional, but much of the wetland vegetation still reads as a green grass sward in these captures. Plant artwork and pool-bank detail are not equally distinctive in every setting.

# Peatland and small ponds

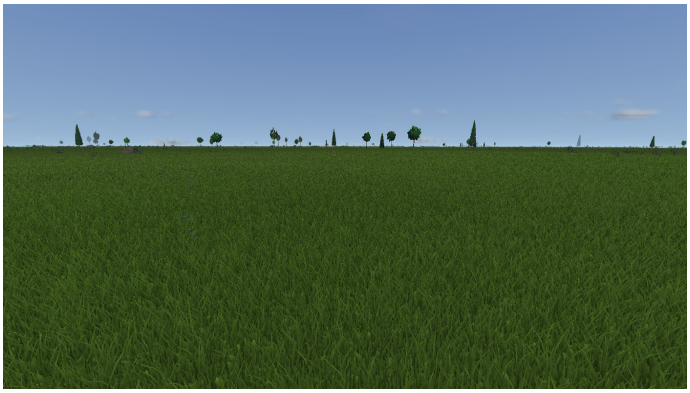
Small standing water now has a real basin, not just a decorative water plane laid over dry ground.

## Peatland

Organic wet-ground controls favour moss/sedge cover and low woody density. This site contains five nearby pools. Its current appearance is close to open marsh; the distinction is stronger in the ground and cover data than in the artwork.



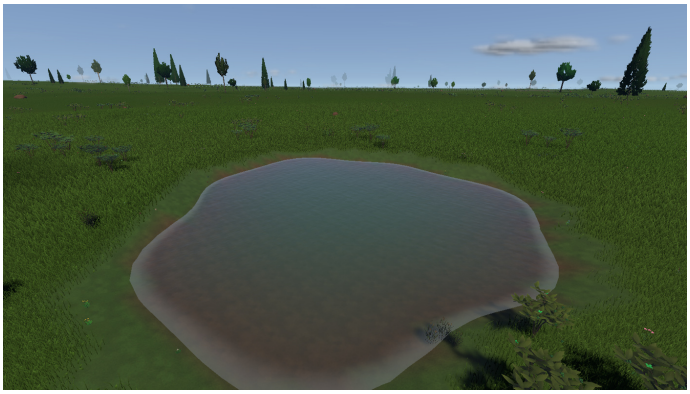
Wide view



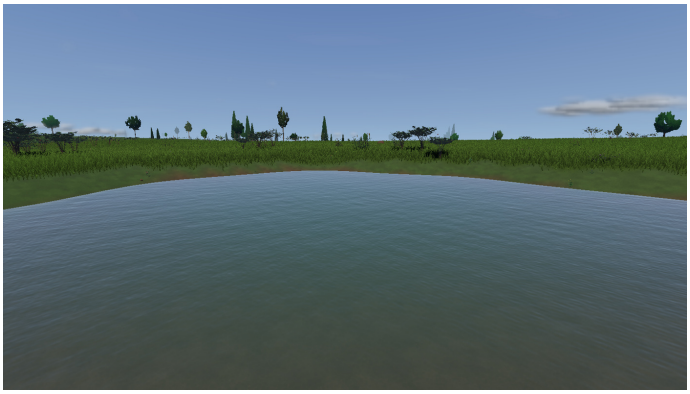
Ground detail

## Enclosed ponds

Candidate depressions are surveyed for a gentle, enclosed rim and given an actual depressed bed. This example has a roughly 14.6 m radius and is 1.71 m deep at the sampled centre. Drinking, swimming and visible water use the same basin.



Wide view



Ground detail

## A small-water mosaic

Sparse deeper ponds are supplemented by finer shallow pools in suitable wetlands. Basin collars cannot overlap; dry banks, beds and water surfaces remain consistent across terrain chunks. Many outlines still look fairly regular in these examples.

## Static water bodies for now

Seasonal drying, freezing/thawing and a local water budget are not implemented. Pond density and shape remain tunable game parameters, rather than a claim to reproduce Earth's exact distribution.

# Uplands and cold country

Rock type, exposed substrate and cold-ground history add variation beyond the desert and wetland cases.

## Limestone uplands

Broken upland relief and exposed substrate contrast with soil-bearing, vegetated gullies. This illustrates regional ground character; it does not add caves, limestone towers or a separate landmark catalogue.



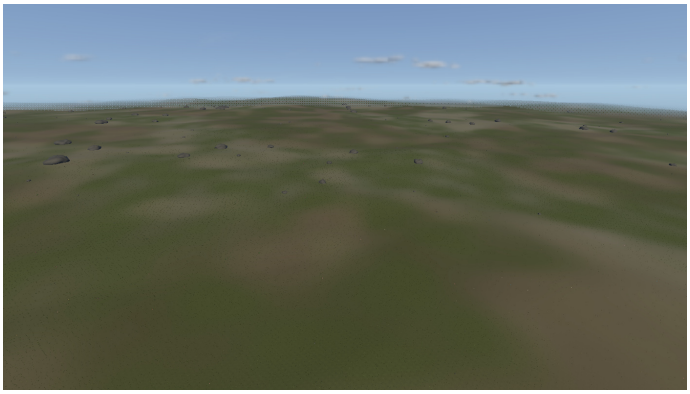
Wide view



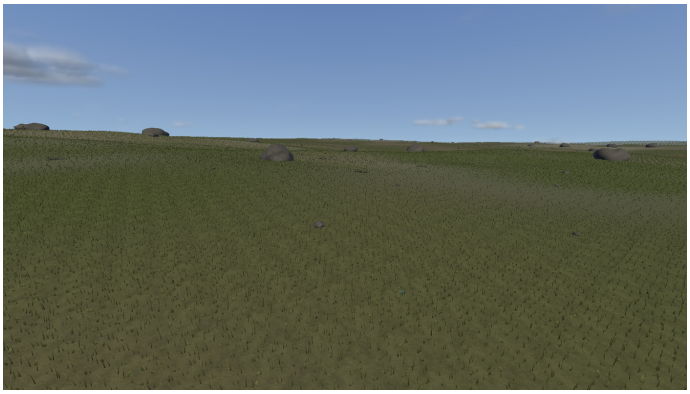
Ground detail

## Glacial lowlands

Open, low-relief ground carries coarse stones and short patchy cover. This is exposed glacially influenced ground, not active ice cover. Permafrost alone no longer labels a region as a glacial landscape.



Wide view



Ground detail

## Scope and cost

The broad world generator is retained. Local terrain/material controls and bounded pond caches add detail on demand; there is no worldwide per-pond tick simulation. Other controls include rocky/alpine ground, tundra, icefields and tidal wet settings. Waterfalls, volcanoes and other landmark features remain outside this update.

Capture record: ten native cases / 84 checks at the verified regional checkpoint (3242861), before the later tree-art update. These are reused, hash-checked captures, not new tests of the current build. Source record: landscape-regions-20261103.json. No gameplay changes were made for this PDF.